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① Efficiency (total value over outcomes is maximized)

- IC
  - BIC
  - DIC

- IR
  - Ex-Ante
  - Interim
  - Ex-Post

- Budget-Balanced
  - $\sum t(\cdot) = 0$ (strict)
  - $\sum t(\cdot) \geq 0$ (weak)

↳ can relax to ex-ante $E[\sum t(\cdot)] = 0$

② VCG Families:

↳ DIC
↳ Efficient $\Big\}$ always (because DIC ⇒ efficient)

* Pivotal VCG Mechanism
↳ if $\nexists$ no negative externalities: $v_i(\cdot, \theta) \geq 0 \ \forall i, \forall \theta$
+ no single agent effect ⇒ weakly budget balanced.
↳ monotonic ⇒ dropping agent doesn't change outcome set.

if no neg. extern, single-agent, monotonic, all properties are satisfied

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Question 5: HW4

a) $\sum_{i=1}^{2} t_i(v_i) = c$ if $x(v_1, v_2) = 1$ (Budget balance)

$= 0$ else

$u(v_i) \geq u(\hat{v}_i) \Rightarrow u_i(v_i) = v_i x(v_1, v_2) - t_i(v_1, v_2) x(v_1, v_2)$

$\geq v_i x(\hat{v}_1, v_2) - t_i(\hat{v}_1, v_2) x(\hat{v}_1, v_2)$

$\forall v_1, v_2, \hat{v}_1, \hat{v}_2$

use envelope thm to calc. payoffs of i w/ v_i
get to here:

$U_1(v_1, v_2) = v_1 x(v_1, v_2) - t_1(v_1, v_2) x(v_1, v_2)$

$U_1(0, v_2) + \int_0^{v_1} x(s, v_2) ds = U_1(v_1, v_2)$

$U_2(v_1, v_2) = u_2(v_1, 0) + \int_0^{v_2} x(v_1, s) ds$

b) $u_1 + u_2 = E[\text{surplus}]$

c) derivate twice

$$u_1(v_1, v_2) + u_2(v_1, v_2) = \ldots \underline{\underline{x(v_1, v_2)}}$$

derivate
cross-partial

d) work w/ c

$$\frac{\partial f}{\partial x \partial y} = 0 \Rightarrow f(x,y) = G(x) + H(y)$$